Chapter 12

Discount

Discount is defined as the amount of rebate given on a fixed price (called as marked price) of an article. It is given by merchants/ shopkeepers to increase their sales by attracting customers.

$$\text{Discount} = \text{Marked Price} - \text{Selling Price}$$

Marked Price (List Price)

The price on the label of an article/product is called the marked price or list price. This is the price at which product is intended to be sold. However, there can be some discount given on this price and actual selling price of the product may be less than the marked price. It is generally denoted by MP.

Basic Formulae Related to Discount

1. $\text{Discount\%} = \dfrac{\text{Marked Price - Selling Price}}{\text{Marked Price}} \times 100$
2. $\text{Discount\%} = \dfrac{\text{Discount}}{\text{Marked Price}} \times 100$
3. $\text{Selling Price} = \text{Marked Price} \times \left(1 - \dfrac{r}{100}\right)$ where, r% is the rate of discount allowed
4. Selling price = Marked price - Discount

Note: Discount is always calculated with respect to marked price of an article

12.1 Successive Discount

When a series of discounts (one after the other) are allowed on marked price of an article, then these discounts are called successive discounts.

Let $r_1\%, r_2\%, r_3\%$ … be the series of discounts on an article with marked price of ₹P, then the selling price of the article after all the discounts is given as

$$\text{Selling Price } = P \times \left(1 - \frac{r_1}{100}\right) \times \left(1 - \frac{r_2}{100}\right) \times \left(1 - \frac{r_3}{100}\right) \times \ldots$$

Ex. 1 An item is sold for ₹680 by allowing a discount of 15% on its marked price. Then what is the marked price of item.

Sol. Selling price of an item = ₹680

And rate of discount, r = 15%

$$\therefore \quad \text{Selling price } = \text{ Marked price } \times \left(1 - \frac{r}{100}\right)$$

$$\implies \quad 680 = \text{MP}\left(1 - \frac{15}{100}\right) = \text{MP}\left(\frac{100-15}{100}\right)$$

$$\implies \quad 680 = \text{MP}\frac{85}{100}$$

$$\therefore \quad \text{Marked Price } = \frac{680 \times 100}{85} = ₹800$$

Ex. 2 A shopkeeper on the eve of Diwali allowed a series of discount on television sets. Find the selling price of a television set, if the marked price of television is ₹1000 and successive discounts are 10% and 5%.

Sol. Selling price of a television = Marked Price $\times \left(1 - \frac{r_1}{100}\right) \times \left(1 - \frac{r_2}{100}\right)$

Here, Marked Price = ₹1000, $R_1 = 10\%$ and $R_2 = 5\%$

$$\therefore \quad \text{Selling Price } = 1000 \times \left(1 - \frac{10}{100}\right) \times \left(1 - \frac{5}{100}\right)$$

$$= 1000 \times \left(\frac{100-10}{100}\right) \times \left(\frac{100-5}{100}\right)$$

$$= 1000 \times \frac{90}{100} \times \frac{95}{100} = ₹\,855$$

12.2 Fast Track Techniques to solve the QUESTIONS

Technique 1

Single discount equivalent to two successive discounts $r_1\%$ and $r_2\%$

$$\text{Single Equivalent Discount} = \left(r_1 + r_2 - \frac{r_1 \times r_2}{100}\right)\%$$

Ex. 3 What will be a single equivalent discount for successive discounts of 10% and 5% on marked price of an article?

Sol. Given, $r_1 = 10\%$ and $r_2 = 5\%$

$$\text{Single equivalent discount} = \left(r_1 + r_2 - \frac{r_1 \times r_2}{100}\right)\%$$
$$= \left(10 + 5 - \frac{10 \times 5}{100}\right)\% = \left(15 - \frac{50}{100}\right)\%$$
$$= (15 - 0.5)\% = 14.5\%$$

Technique 2

Single discount equivalent to three successive discounts $r_1\%$, $r_2\%$ and $r_3\%$.

$$\text{Single Equivalent Discount} = \left[1 - \left(1 - \frac{r_1}{100}\right)\left(1 - \frac{r_2}{100}\right)\left(1 - \frac{r_3}{100}\right)\right] \times 100\%$$

Ex. 4 What is the single equivalent discount for successive discounts of 10%, 20% and 15% on marked price for motor bike?

Sol. Given, $r_1 = 10\%$, $r_2 = 20\%$ and $r_3 = 15\%$

Single equivalent discount for three successive discounts

$$= \left[1 - \left(1 - \frac{r_1}{100}\right)\left(1 - \frac{r_2}{100}\right)\left(1 - \frac{r_3}{100}\right)\right] \times 100\%$$
$$= \left[1 - \left(1 - \frac{10}{100}\right)\left(1 - \frac{20}{100}\right)\left(1 - \frac{15}{100}\right)\right] \times 100\%$$
$$= \left[1 - \left(\frac{90}{100}\right)\left(\frac{80}{100}\right)\left(\frac{85}{100}\right)\right] \times 100\% = \left[1 - \frac{9}{10} \times \frac{8}{10}\frac{17}{20}\right] \times 100\%$$
$$= \left[\frac{1000 - 612}{1000}\right] \times 100\% = \frac{388}{1000} \times 100\% = 38.8\%$$

Technique 3

If a shopkeeper wants a profit of $R\%$ after allowing a discount of $r\%$, then

$$\text{Marked Price(MP) of the item} = CP\left(\frac{100 + R}{100 - r}\right)$$

OR

$$\text{Cost Price(CP) of the item} = MP\left(\frac{100 - r}{100 + R}\right)$$

Ex. 5 The marked price of a bicycle is ₹1100. A shopkeeper allows a discount of 10% and get a profit of 10%. Find the cost price of the bicycle.

Sol. Let CP of the bicycle= ₹ x

$$\text{Then, SP of the bicycle } = \frac{110x}{100} - \frac{11x}{10}$$

$$\text{Now, MP of the bicycle} = ₹\ 1100$$

$$\therefore \quad \text{SP of the bicycle} = \frac{1100 \times 90}{100} = ₹\ 990$$

According to the question,

$$\frac{11x}{10} = 990$$

$$\Longrightarrow \quad x = \frac{990 \times 10}{11} = ₹\ 900$$

Fast Track Method

Here, $MP = ₹\ 1100,\ r = 10\%$ and $R = 10\%$

$$\therefore \quad CP = MP\left(\frac{100 - r}{100 + R}\right)$$

$$= 1100 \times \frac{100 - 10}{100 + 10} = \frac{1100 \times 90}{110} = ₹\ 900$$

Technique 4

A merchant fixes the marked price of an article in such a way that after allowing a discount of r%, he earns a profit of R%. Then,

$$\text{Marked Price of the article is } \left(\frac{r + R}{100 - r} \times 100\right)\% \text{ more than its Cost Price}$$

Ex. 6 A shopkeeper allows a discount of 10% on the marked price of calculator. then by what per cent higher than cost price, should marked the price, so as to gain 20% on selling it at the discount?

Sol. Let the cost price of article be ₹100

Then, selling price should be 120% of 100

$$= \frac{120 \times 100}{100} = ₹\ 120$$

Now, selling price of the article should be 10% lower than marked price because of the discount of 10%

$$\therefore \quad \text{Marked Price } = \frac{100 \times\ SP}{100 - 10}$$

$$= \frac{100 \times 120}{90} = \frac{400}{3}$$

$\therefore$ Required percentage at which article is marked higher than cost price.

$$= \frac{MP - CP}{CP} \times 100\%$$

$$= \frac{\frac{400}{3} - 100}{100} \times 100 = \frac{(400 - 300)}{3 \times 100} \times 100 = \frac{100}{3} = 33\frac{1}{3}\%$$

Fast Track Method

Here, $r = 10\%$ and $R = 20\%$

$$\text{Required percentage} = \left(\frac{r + R}{100 - r} \times 100\right)\% = \frac{10 + 20}{100 - 10} \times 100$$

$$= \frac{30 \times 100}{90} = \frac{100}{3} = 33\frac{1}{3}\%$$

Technique 5

If a shopkeeper allows a discount of $r_1\%$ on an article and marked price of the article is $r\%$ more than the cost price, then

$$\text{Profit or Loss percent} = \frac{r \times (100 - r_1)}{100} - r_1$$

Note: Positive value shows a profit, while negative value shows loss

Ex. 7 A shopkeeper marked the price 10% more than its cost price. If he allows a discount of 20%, then find his loss per cent.

Sol. Here, $r_1 = 20\%$ and $r = 10\%$

$\therefore$ Loss or profit percent $= \frac{r \times (100 - r_1)}{100} - r_1 = \frac{10 \times (100 - 20)}{100} - 20$

$$= \frac{10 \times 80}{100} - 20 = -12\%$$

$\therefore$ Loss percent $= 12\%$